OpenClaw Voice Dialogue Interaction

1. Course Content

Course Overview

OpenClaw Voice Dialogue Interaction is a **full-duplex natural language voice interaction solution** designed specifically for the A1 Pro Ackermann embodied intelligence robot. By bridging local high-performance voice processing capabilities (microphone array hardware wake-up, Silero-VAD voice activity detection, real-time speech recognition ASR, speech synthesis TTS) with the OpenClaw embodied large model brain and the ROS 2 MCP (Model Context Protocol) chassis/arm control framework, users can engage in intelligent chat, common sense Q&A, and drive robot chassis movement and arm spatial grasping operations through natural voice dialogue - without the need for keyboard or web text input.

Learning Objectives

- Understand the system architecture of the OpenClaw voice interaction loop and the collaboration mechanism of each node
- Master the method to start the chassis integrated service node `mcp_service.launch.py` that includes OpenClaw bridging
- Master the method to start the speech recognition and wake-up node `asr_detect`, and be proficient in using **hardware wake-up word** and **terminal spacebar wake-up**
- Be able to issue embodied commands such as chat Q&A, chassis movement, and arm spatial operations to the robot through natural voice
- Understand the configuration modifications for the core parameter configuration file `common.yaml` and common troubleshooting tips

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2. System Architecture and Working Principle

Core Interaction Timing Flow

1. **Wake-up and Interrupt**: User calls the hardware wake-up word **"Hi, yahboom"**, `asr_detect` captures the wake-up event, publishes the `/wakeup_signal` topic to interrupt the currently playing TTS broadcast, and plays a response prompt tone ("I'm here");
2. **Speech-to-Text (ASR)**: User speaks the command, Silero VAD module detects human voice and automatically closes the audio stream after the user finishes speaking, ASR engine precisely converts it to text and publishes to `/asr_result`;
3. **LLM Reasoning and Tool Calling**: `openclaw_bridge` delivers the text to OpenClaw Gateway, the large model analyzes user intent:
    - If it belongs to **general conversation/common sense Q&A**: directly generate response text;
    - If it belongs to **robot embodied control**: the large model directly calls robot operation tools provided by `mcp_service` through the MCP protocol (for example, making the car move forward, controlling arm movement, etc.);

4. **Speech Synthesis and Feedback (TTS)**: The response content is printed in real-time streaming via `openclaw_bridge` terminal, while simultaneously calling the TTS engine to synthesize speech and deliver it to the speaker for playback, achieving an integrated voice-action real interaction.

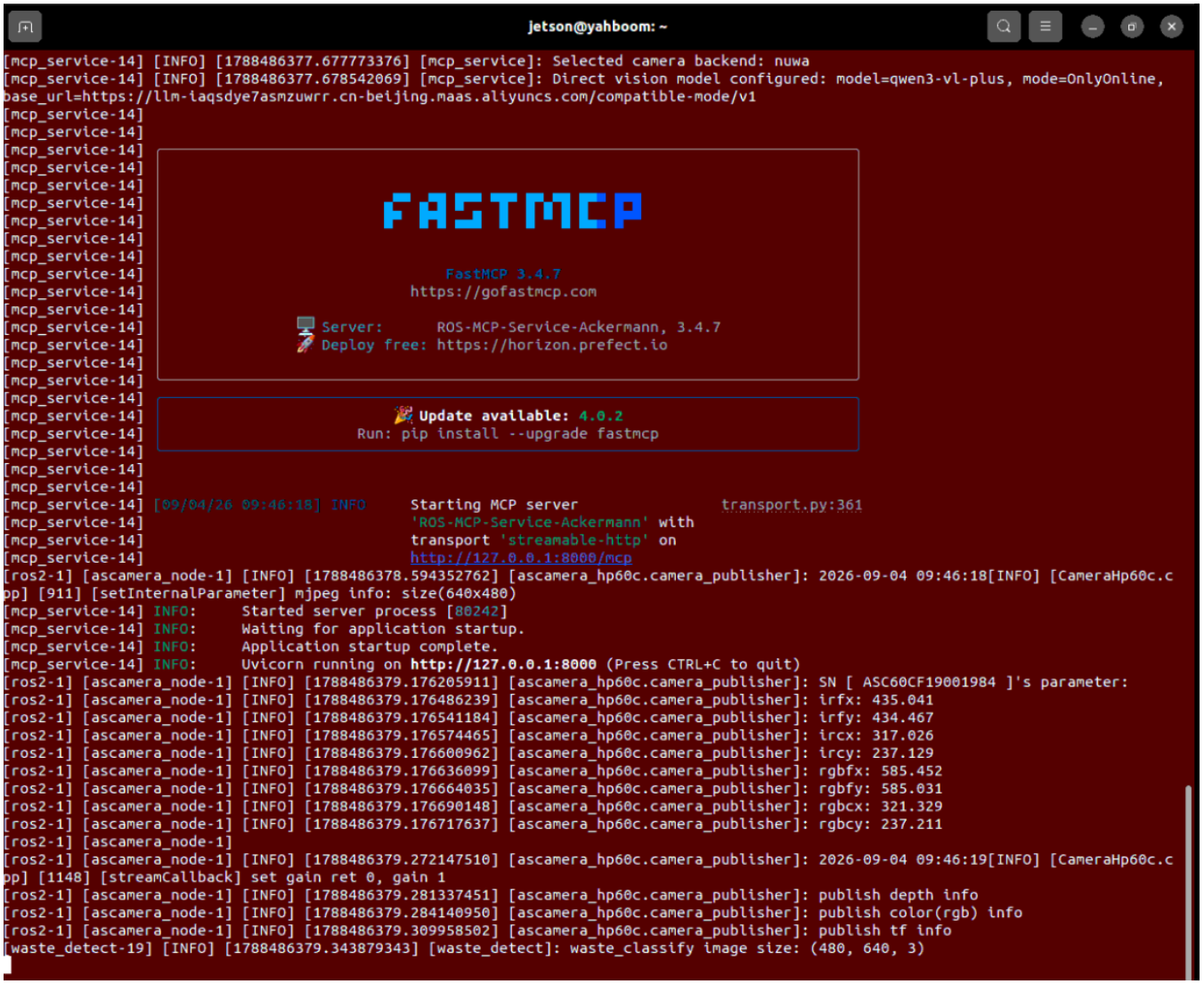
3. Voice Interaction Startup Steps

- Start core nodes such as odometry, TF transformation, arm assistance, camera nodes, and MCP service:

```
ros2 launch dofbot_pro_bringup mcp_service.launch.py
```

- If you need voice broadcast responses, you can start the program with the `launch_openclaw_bridge` parameter, otherwise you can use the basic command above:

```
ros2 launch dofbot_pro_bringup mcp_service.launch.py   launch_openclaw_bridge:=true
```



- Run the voice dialogue interaction with the following command:

```
ros2 run ackermann_mcp asr_detect
```

```
jetson@yahboom:~/yahboomcar_ros2_ws/software/vioce_toolbox_lib$ ros2 run ackermann_mcp asr_detect

ASR supplier: aliyun
[INFO] [1788487956.600357547] [asr_detect]: enable_wakeup_voice: True, enable_keybord_wakeup: False, en
able_kws_wakeup: True
[INFO] [1788487957.180189913] [asr_detect]:  ASR Engine: paraformer-realtime-v2 initialized successfull
y
[voice_toolbox] v1.0.0 loaded, copyright Yahboom Technology
[INFO] [1788487957.428767489] [asr_detect]: VAD engine initialized successfully
[kws_mic] Serial /dev/myspeech opened successfully
```

After waking up by calling the wake-up word "**Hi, yahboom**", the indicator light turns on, and you can speak the content you want to dialogue about:

```
jetson@yahboom:~/yahboomcar_ros2_ws/software/vioce_toolbox_lib$ ros2 run ackermann_mcp asr_detect

ASR supplier: aliyun
[INFO] [1788487956.600357547] [asr_detect]: enable_wakeup_voice: True, enable_keybord_wakeup: False, en
able_kws_wakeup: True
[INFO] [1788487957.180189913] [asr_detect]:  ASR Engine: paraformer-realtime-v2 initialized successfull
y
[voice_toolbox] v1.0.0 loaded, copyright Yahboom Technology
[INFO] [1788487957.428767489] [asr_detect]: VAD engine initialized successfully
[kws_mic] Serial /dev/myspeech opened successfully
recording...
[✓] 给我讲一个笑话。
```

Once successfully woken up, you can chat with OpenClaw as usual

> **Tip**
>
> Whether waking up via voice call or pressing the spacebar, when you hear the robot prompt tone, you can directly speak your command naturally.

Typical Dialogue and Control Cases

Case 1: Intelligent Common Sense and Chat Q&A

- **User Question**: "Introduce yourself, what functions do you have?"

Case 2: Chassis Motion Control Command

- **User Command**: "Move forward 0.5 meters."

Case 3: Arm Posture and Action Control

- **User Command**: "Have the arm say hello to me."

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4. Parameter Configuration and Personalized Tuning

The core parameters of the voice interaction system are defined in the following configuration file:

```
$HOME/yahboomcar_ros2_ws/dofbot_pro_ws/src/dofbot_pro_bringup/config/common.yaml
```

Open this file with an editor to modify the following two configuration sections as needed:

4.1 `asr_detect` Speech Recognition Node Configuration

```yaml
asr_detect:
  ros__parameters:
    enable_wakeup_voice: True      # Whether to play a prompt tone ("I'm here") on wake-up
    enable_keybord_wakeup: True     # Whether to enable keyboard spacebar wake-up function
    enable_kws_wakeup: True         # Whether to enable hardware serial voice wake-up
function
    language: "zh"                  # Language setting: zh for Chinese, en for English
    use_oline_asr: True             # Whether to use cloud online speech recognition (True
has higher recognition rate)
    asr_supplier: "aliyun"          # Speech recognition service provider: aliyun or xunfei
    oline_asr_model: "paraformer-realtime-v2" # Aliyun online speech recognition model
    aliyun_api_key: "your-api-key"  # Aliyun API Key (can also configure environment
variable ALIYUN_API_KEY)
    mic_serial_port: "/dev/myspeech"# Microphone hardware serial device name
    no_speech_timeout: 5.0          # Maximum waiting time for no sound input after wake-up
(seconds)
```

4.2 `openclaw_bridge` Bridge Node Configuration

```yaml
openclaw_bridge:
  ros__parameters:
    enable_voice_reply: True        # Whether to enable TTS voice broadcast reply (set to
False for terminal text only)
    language: "zh"                  # Broadcast language: zh for Chinese, en for English
    openclaw_base_url: "http://127.0.0.1:18789/v1" # OpenClaw Gateway OpenAI-compatible
interface address
    openclaw_api_key: "yahboom"     # OpenClaw gateway connection token
    openclaw_session: "robot-session-001" # Conversation session ID
    use_oline_tts: True             # Whether to use online cloud TTS
    tts_supplier: "aliyun"          # Speech synthesis service provider: aliyun / baidu /
xunfei
    tongyi_tts_model: "cosyvoice-v1" # Aliyun speech synthesis model
    voice_tone: "longxiaochun"      # Voice timbre (such as intellectual female voice, child
voice, young male voice, etc.)
    tts_playback_timeout: 120.0     # Maximum timeout for single audio playback
```

> **Note**
>
> After modifying the `common.yaml` file, you must **restart the corresponding ROS 2 node** for the parameters to take effect.

5. Common Problems and Solutions

5.1 Starting `asr_detect` Shows Cannot Open Serial Port `/dev/myspeech`

Symptoms: Terminal reports `Cannot open serial port /dev/myspeech, please check voice wakeup module connection`.

Possible causes and solutions:

- **Hardware voice module not connected**: If the hardware speech recognition module is not yet plugged in, this is a normal prompt. In this case, the program will automatically degrade; you can perform keyboard wake-up interaction by **pressing the spacebar in the terminal**, and the functionality will not be affected;
- **Insufficient serial port permissions**: Check if you have access permissions, run `ls -l /dev/myspeech`; if permissions are insufficient, execute `sudo chmod 777 /dev/myspeech` or add the current user to the `dialout` group.

5.2 No Recognized Text Output in Terminal After Speaking

Symptoms: After wake-up, speaking but the terminal does not print recognized text.

Possible causes and solutions:

- **Microphone input volume too low or environment noisy**: Speak clearly closer to the microphone, or check the USB microphone recording gain via system `alsamixer`;
- **Online ASR service configuration abnormal**: If using `aliyun` or `xunfei` online ASR, check if the network connection is normal and if the API-KEY is valid.

5.3 Terminal Recognizes Text, But OpenClaw Has No Response or Reports Connection Failed

Symptoms: `asr_detect` outputs text normally, but terminal 1's `openclaw_bridge` reports `OpenClaw Request failed: Connection refused`.

Possible causes and solutions:

- **OpenClaw gateway not running**: Execute `openclaw gateway status` to check gateway status; if not started, execute `openclaw gateway restart` to restart the gateway;
- **Port or address mismatch**: Confirm whether `openclaw_base_url` in `common.yaml` matches the gateway's actual listening address and port (default `http://127.0.0.1:18789/v1`).